

Experiment 1 — Simulating Radioactive Decay Through Dice

Roll (t)	No. of Atoms Decayed (Each Trial)	No. of Atoms Decayed (Cumulative)	No. of Atoms Decayed (Theoretical)	N(t) Theoretical	N(t) Experimental	ln N(t) Experimental
0	0	0	0	99	99	4.6
1	15	15	15	84	84	4.43
2	15	30	28	69	71	4.23
3	11	41	39	58	60	4.06
4	10	51	48	48	51	3.87
5	11	62	56	37	43	3.61
6	3	65	63	34	36	3.53
7	2	67	68	32	31	3.47
8	6	73	73	26	26	3.26
9	5	78	77	21	22	3.04
10	4	82	80	17	19	2.83
11	3	85	83	14	16	2.64
12	1	86	86	13	13	2.56
13	2	88	88	11	11	2.4
14	2	90	89	9	10	2.2
15	0	90	91	9	8	2.2
16	1	91	92	8	7	2.08
17	1	92	93	7	6	1.95
18	2	94	94	5	5	1.61
19	0	94	95	5	4	1.61
20	1	95	95	4	4	1.39
21	0	95	96	4	3	1.39
22	3	98	96	1	3	0
23	1	99	97	0	2	

Radioactive Decay Simulation Using Dice

